

RUMOI, Japan

WMO: 474060

Lat: 43.9461N

Lon: 141.6319E

Elev: 25

StdP: 101.02

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-10.5	-15.7	1.0	-10.8	-14.0	1.1	-9.1	15.3	-3.0	14.0	-3.7	5.5	110	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	27.0	22.5	25.5	21.7	24.3	21.1	23.4	25.9	22.6	24.7	21.8	23.8	4.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.3	25.0	21.8	16.5	24.3	21.0	15.7	23.5	69.9	25.9	67.0	24.6	64.1	23.9	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	12.0	10.7	-16.2	30.4	2.3	1.7	-17.9	31.6	-19.2	32.6	-20.6	33.6	-22.3	34.8
			-16.5	24.6	2.3	1.2	-18.2	25.5	-19.6	26.2	-20.9	26.8	-22.6	27.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-4.4	-3.9	-0.3	5.4	11.2	15.5	19.7	21.2	17.3	11.0	4.3	-1.6
	DBStd	9.40	2.95	3.25	3.10	3.27	3.47	2.91	2.56	2.45	3.06	3.05	3.63	3.01
	HDD10.0	1883	446	390	318	144	26	1	0	0	24	173	359	
	HDD18.3	3948	704	623	577	388	222	93	14	5	56	228	420	617
	CDD10.0	1158	0	0	0	6	63	165	302	346	218	55	3	0
	CDD18.3	182	0	0	0	0	1	8	57	92	23	0	0	0
	CDH23.3	501	0	0	0	0	6	16	124	292	64	0	0	0
	CDH26.7	59	0	0	0	0	1	17	33	8	0	0	0	0
Wind	WSAvg	5.1	6.3	6.0	5.6	4.9	4.4	3.6	3.5	3.9	4.6	5.7	6.5	6.8
Precipitation	PrecAvg	1168	102	73	55	49	63	59	114	122	150	123	137	121
	PrecMax	1466	144	134	93	100	113	110	227	220	364	221	226	194
	PrecMin	902	59	35	16	12	18	24	26	41	60	46	85	63
	PrecStd	145	26	25	20	23	22	19	56	49	77	44	41	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	6.0	10.2	19.1	23.7	25.2	28.8	29.5	27.6	20.7	15.9	8.8
		MCWB	2.3	3.2	5.6	11.1	15.0	18.5	23.2	24.1	22.5	15.8	12.0	6.8
	2%	DB	2.4	3.7	7.4	15.1	20.5	22.8	26.3	27.4	25.0	18.9	13.6	5.9
		MCWB	0.6	1.7	4.2	9.4	14.0	17.8	22.0	23.1	21.1	15.0	10.6	3.8
	5%	DB	1.1	2.4	5.9	12.5	18.3	20.9	24.5	26.0	23.5	17.6	11.9	4.3
		MCWB	-0.4	0.5	3.4	8.1	13.2	17.2	21.3	22.4	20.0	13.9	9.1	2.3
	10%	DB	0.0	1.0	4.5	10.5	16.3	19.4	23.3	24.9	22.2	16.3	10.4	2.9
		MCWB	-1.6	-0.7	2.3	7.1	12.4	16.4	20.8	21.8	19.0	12.9	7.7	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.6	6.4	12.1	16.4	20.2	24.2	24.7	23.2	17.2	12.6	6.7
		MCDB	3.9	5.4	9.1	16.9	21.1	23.3	27.8	28.0	26.3	19.4	15.3	8.6
	2%	WB	0.8	1.9	4.8	10.1	14.9	18.6	22.8	23.7	22.0	15.7	11.0	4.1
		MCDB	1.9	3.5	6.7	13.9	18.7	21.4	24.9	26.4	24.3	18.1	13.2	5.6
	5%	WB	-0.2	0.7	3.6	8.8	13.9	17.7	22.0	23.0	20.6	14.4	9.5	2.5
		MCDB	1.0	2.2	5.7	11.8	17.3	20.0	24.0	25.2	22.7	16.9	11.7	4.1
	10%	WB	-1.5	-0.6	2.5	7.5	12.9	16.8	21.1	22.3	19.5	13.2	7.9	1.1
		MCDB	-0.2	0.9	4.4	10.2	15.8	18.9	22.8	24.4	21.8	15.7	10.1	2.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	6.1	7.1	7.4	6.1	5.5	6.1	7.6	7.6	6.0	5.1
		MCDBR	6.3	7.1	8.1	11.0	10.8	8.5	7.3	7.1	8.2	9.1	8.9	6.7
		MCWBR	5.6	6.2	5.9	6.5	5.7	4.6	3.9	3.9	4.8	6.0	7.0	5.8
	5% WB	MCDBR	6.3	6.9	7.6	10.0	9.5	7.2	6.3	6.0	7.0	8.2	8.4	6.8
		MCWBR	5.9	6.3	6.0	6.3	5.3	4.2	3.5	3.5	4.5	5.9	7.1	6.1
Clear-Sky Solar Irradiance	taub	0.352	0.405	0.465	0.525	0.538	0.522	0.500	0.465	0.414	0.414	0.398	0.371	
	taud	2.021	1.885	1.805	1.792	1.858	1.984	2.113	2.231	2.318	2.156	2.092	1.998	
	Ebn at Noon	720	742	751	744	751	766	778	793	808	737	666	646	
	Edh at Noon	110	152	188	206	199	176	153	132	112	115	103	103	
All-Sky Solar Radiation	RadAvg	1.08	1.87	3.12	4.54	5.19	5.15	4.79	4.53	3.73	2.46	1.28	0.85	
	RadStd	0.10	0.22	0.32	0.44	0.43	0.55	0.57	0.32	0.32	0.17	0.16	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page